

Jia-Lai Kang (康 嘉来)

Curriculum Vitae



➤ Education

- 2020-2026 Ph.D. in Astronomy, University of Science and Technology of China
Thesis: Observational Studies of the X-ray Corona in Active Galactic Nuclei
Supervisor: Prof. Jun-Xian Wang
- 2024-2025 Visiting student of Durham University
Supervisor: Prof. Chris Done & Martin Ward
- 2016-2020 B.S. in Astronomy, University of Science and Technology of China (USTC)
Thesis: Hard X-ray spectral analysis of Swift/BAT-selected Seyfert galaxies
Supervisor: Prof. Jun-Xian Wang

➤ Awards

- 2026 Outstanding Graduates, USTC
- 2026 CAS President's Award (Excellence Prize), CAS
- 2026 Outstanding Doctoral Dissertations, USTC
- 2024 Youth Students Program in Basic Science (grant No. 123B2042: 300,000 CNY), NSFC
- 2023 China National Scholarship
- 2021 China National Scholarship
- 2020 Outstanding Graduates, USTC

➤ First-author Publications

- 1) **Kang, J.-L.**, & Wang, J.-X., 2026, "Sudden emergence of a low-frequency hard X-ray lag in the Seyfert 1 galaxy Mrk 1044", accepted by ApJ
- 2) **Kang et al.** 2025b, "Disk-jet coupling across the spectral transition in supermassive black holes", MNRAS, Volume 543, Issue 1, pp.709–721
- 3) **Kang et al.** 2025a, "Systematic collapse of the accretion disc in AGN confirmed by UV photometry and broad-line spectra", MNRAS, Volume 538, Issue 1, pp.121-131
- 4) **Kang et al.** 2024, "On joint analysing XMM-NuSTAR spectra of active galactic nuclei", JUSTC, 54 (7): 0702–1–0702–16
- 5) **Kang et al.** 2023b, "What can be learnt from a highly informative X-ray occultation event in NGC 6814? A marvellous absorber", MNRAS, Volume 525, Issue 2, pp.1941-1952
- 6) **Kang, J.-L.**, & Wang, J.-X., 2023a, "Hidden biases in flux-resolved X-ray spectroscopy", MNRAS, Volume 519, Issue 3, pp.3635-3642
- 7) **Kang, J.-L.**, & Wang, J.-X., 2022, "The X-ray coronae in NuSTAR bright active galactic nuclei", ApJ, 929, 141
- 8) **Kang et al.** 2021, "Distinct high-energy cutoff variation patterns in two Seyfert galaxies", MNRAS, Volume 502, Issue 1, pp.80-88

9) **Kang et al.** 2020, “NuSTAR hard X-ray spectra of radio galaxies”, ApJ, 901, 111

➤ **Talks**

- 2024.12 “Accretion Alliance” Durham, UK
Disk-jet coupling across the spectral transition in supermassive black holes
- 2024.09 “New Results in X-ray astronomy / NewAthena UK meeting” Sussex, UK
Systematic collapse of the accretion disc in AGN (shared with Scott Hagen)
- 2023.06 “The Restless Nature of AGN: 10 years later” Napoli, Italy
On the coronal temperature and its variability
- And more than 10 talks in China

➤ **Instruments and Programming**

- Extensive experience in spectral and timing analysis of X-ray data from major observatories, particularly NuSTAR, XMM-Newton (EPIC and RGS), and eROSITA.
- Proficient in the analysis of multi-band survey data, including radio (VLASS, FIRST, ASKAP), infrared (WISE), optical (SDSS, DESI, HSC-SSP, KiDS), ultraviolet (GALEX), and X-ray (eROSITA).
- Experienced in large optical spectroscopic surveys, including the use of PyQSOFit to measure spectral properties of SDSS and DESI quasars.
- Skilled in Python and Shell scripting for large-scale data processing and analysis on computing clusters.